A small minimal resolution over the Steenrod algebra, computed by Gröbner bases

Trace of the `groebner` crate (branch `groebner_lin`)

Abstract

We follow, one reduction at a time, the computation of the minimal free resolution $\cdots \to C_3 \to C_2 \to C_1 \to C_0 \to \mathbb{F}_2$ of the ground field over the mod 2 Steenrod algebra $\mathcal{A}$ by the `groebner` crate, a port of Weinan Lin's Gröbner basis method. The range is homological degree $s \leq 3$ and stem $t - s \leq 4$, so internal degrees $t \leq 7$. Every formula below is taken from the program's own data: the critical pairs it formed, the elements it reduced, and the bases and differentials it produced. Only the bookkeeping of the reduction steps was replayed outside the program, and the replay was checked to reproduce the program's results exactly.

1 Setting

The algebra. We work in the Milnor basis $\mathrm{Sq}(R) = \mathrm{Sq}(r_1, r_2, \dots)$ of $\mathcal{A}$, writing $\mathrm{Sq}^r = \mathrm{Sq}(r)$. Write $r_n = \sum_i d_{i,n} 2^i$ in binary. The digit $d_{i,n}$ records the power $\xi_n^{2^i}$ of the dual basis, of degree $2^i(2^n - 1)$, and we record $\mathrm{Sq}(R)$ by the monomial

$$x^R = \prod_{d_{i,n}=1} x_{in}$$

in the associated graded algebra of the May filtration

$$E^0\mathcal{A} = \bigotimes_{i \geq 0,\, n \geq 1} \mathbb{F}_2[x_{in}]/(x_{in}^2).$$

For example $\mathrm{Sq}^3 = \mathrm{Sq}(3) \leftrightarrow x_{01}x_{11}$, $\mathrm{Sq}(0,1) \leftrightarrow x_{02}$ and $\mathrm{Sq}(1,1) \leftrightarrow x_{01}x_{02}$. In $E^0\mathcal{A}$ a monomial is just a set of positions, so divisibility, quotients and least common multiples are inclusion, difference and union of sets. The positions relevant here, in increasing degree, are

$$x_{01}\ (1), \quad x_{11}\ (2), \quad x_{02}\ (3), \quad x_{21}\ (4), \quad x_{12}\ (6), \quad x_{03}\ (7).$$

The monomial order. A term of a free module is $\mathrm{Sq}(R)\, g$ with g a generator. Terms are compared by

1. the generator: the generator added later (of higher degree) comes first;
2. the May weight $\sum_{d_{i,n}=1}(2n-1)$: smaller comes first;
3. the digits read as a binary number, with the position of lowest degree most significant: smaller comes first.

The *leading term* $\mathrm{LT}(x)$ of an element is its first term in this order. Elements are always printed with their leading term first. The point of this order is that the leading term of a product $\mathrm{Sq}(R) \cdot x$ is $x^R \cdot \mathrm{LT}(x)$ computed in $E^0\mathcal{A}$, *whenever that product is non-zero.* Products of leading terms vanish exactly when a position occurs twice, and those are the cases the algorithm treats separately ("singles" below).

Names. The generator of C_0 is ι. Generators of C_s, $s \geq 1$, are named after the classes of $\mathrm{Ext}_{\mathcal{A}}^{s,t}(\mathbb{F}_2, \mathbb{F}_2)$ they represent: $[h_0], [h_1], [h_2] \in C_1$, $[h_0^2], [h_1^2], [h_0 h_2] \in C_2$, and so on. These names are only labels; the program numbers the generators of each C_s in the order they are found.

2 The algorithm

Levels. Level s is a Gröbner basis $b_0^s, b_1^s, \ldots$ of the graph of the differential,

$$\Gamma_s = \{(dy, y) : y \in C_{s+1}\} \subseteq C_s \oplus C_{s+1},$$

for the order on the first component. Each $b_k^s = (x_1 \mid x_2)$ satisfies $dx_2 = x_1$, and its leading term is that of x_1. The first components form a Gröbner basis of $\mathrm{im}(d\colon C_{s+1} \to C_s)$, and pairs whose first component reduces to zero produce elements of the kernel. The module being resolved enters as "level -1": the relations $\mathrm{Sq}^{2^i}\iota = 0$ presenting $\mathbb{F}_2 = \mathcal{A}/\mathcal{A}\{\mathrm{Sq}^1, \mathrm{Sq}^2, \mathrm{Sq}^4, \ldots\}$.

Critical pairs and singles. When an element with leading term $L = x^R g$ joins level s, the program forms

- a *critical pair* with each earlier element b whose leading term L' lies on the same generator: with $M = \mathrm{lcm}(L, L')$ in $E^0\mathcal{A}$, the S-element is $\mathrm{Sq}(M/L') \cdot b + \mathrm{Sq}(M/L) \cdot b_{\mathrm{new}}$, in which the leading terms cancel. The pairs are pruned by the Gebauer–Möller criteria (divisibility and the chain criterion);
- a *critical single* $\mathrm{Sq}(x_{in}) \cdot b_{\mathrm{new}}$ for each position x_{in} dividing L: then $x_{in} \cdot L = 0$ in $E^0\mathcal{A}$, so the leading term of the product is not predicted by the leading terms and the product must be reduced.

A pair or single has the degree of its leading monomial, and it is filed under that degree.

One degree. The program runs through $t = 1, 2, \ldots$. In degree t:

1. Every level hands out its pairs and singles of degree t, and the relations of degree t are taken up.
2. Each of them is reduced: the first component modulo level s, carrying the second component along so that $dx_2 = x_1$ is preserved; then the second component modulo the first components of level $s+1$, i.e. modulo the boundaries $d(C_{s+2})$ known so far, which preserves $dx_2 = x_1$ because $dd = 0$. If the first component has become zero, the second component $z \in C_{s+1}$ is a cycle.
3. Going down from the top level, the results are compared with the elements found earlier in the same degree ("triangulation"), and then sorted:
   - first component non-zero: a **new basis element** of level s;
   - both components zero: **useless**;
   - first component zero, second component $z \neq 0$: z is a cycle that is not a boundary (it is reduced modulo the whole image in degree t), so it becomes d of a **new generator** g of C_{s+2}, and $(z \mid g)$ joins level $s+1$. If $s+2$ exceeds the range, the outcome is **beyond** instead.

Because the levels are handled from the top down, the new elements of level $s+1$ in degree t are known when the cycles of level s are tested, which is what makes the resolution minimal. No linear algebra is ever done: everything is rewriting of sparse sums of terms.

Truncation. A cycle found at level s in degree t is a generator of C_{s+2} in stem $t-s-2$. Since we want stems ≤ 4, level s is only needed for $t \leq s+6$, and pairs of higher degree are never formed.

Running it. The trace was produced by a small driver around the public API of the crate (`Resolution::sphere` with `Bounds { s_max: 3, n_max: 4 }` at $p = 2$, `SteenrodKind::Classical`), which calls `compute_through(t)` for $t = 1, \ldots, 7$ and records every reduction through a `PairObserver`.

3 Step by step

Degree $t = 1$

The work of this degree: the relation $\mathrm{Sq}^1\iota$. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Relation $\mathrm{Sq}^1\iota = 0$. The module relation enters as the element $(0 \mid \mathrm{Sq}^1\iota)$ of level -1, so its C_0 component is a candidate for d of a new generator. It is reduced modulo the image $\mathrm{im}(d\colon C_1 \to C_0)$, i.e. the first components of level 0. No term of it is divisible by a leading term of level 0 from earlier degrees. **Outcome: new generator.** The relation $\mathrm{Sq}^1\iota$ is not a boundary, so C_1 gets a generator $[h_0]$ in degree 1 with $d[h_0] = \mathrm{Sq}^1\iota$. Level 0 gets the element $b_0^0 = (\mathrm{Sq}^1\iota \mid [h_0])$, with leading term $x_{01}\,\iota$. *End of degree 1.* New basis elements: level 0: b_0^0. New generators: $[h_0] \in C_1$.

Degree $t = 2$

The work of this degree: the relation $\mathrm{Sq}^2\iota$; level 0: single. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Level 0, single $x_{01} \cdot b_0^0$. Since x_{01} divides $\mathrm{LT}(b_0^0) = x_{01}\,\iota$ and $x_{in}^2 = 0$ in $E^0\mathcal{A}$, the leading term of $\mathrm{Sq}^1 \cdot b_0^0$ is not the product of the leading terms, and the product has to be reduced:

$$(0 \mid \mathrm{Sq}^1[h_0]).$$

The first component is zero, so the second component $z = \mathrm{Sq}^1[h_0]$ is a cycle in C_1. No term of it is divisible by a leading term of level 1 from earlier degrees. The cycle $\mathrm{Sq}^1[h_0]$ is set aside: cycles are tested once all results of level 0 in this degree are sorted.

The cycles of level 0. There is one. **Outcome: new generator.** The cycle $\mathrm{Sq}^1[h_0]$ is not a boundary, so C_2 gets a generator $[h_0^2]$ in degree 2 with $d[h_0^2] = \mathrm{Sq}^1[h_0]$. Level 1 gets the element $b_0^1 = (\mathrm{Sq}^1[h_0] \mid [h_0^2])$, with leading term $x_{01}\,[h_0]$.

Relation $\mathrm{Sq}^2\iota = 0$**.** The module relation enters as the element $(0 \mid \mathrm{Sq}^2\iota)$ of level -1, so its C_0 component is a candidate for d of a new generator. It is reduced modulo the image $\mathrm{im}(d\colon C_1 \to C_0)$, i.e. the first components of level 0. No term of it is divisible by a leading term of level 0 from earlier degrees. **Outcome: new generator.** The relation $\mathrm{Sq}^2\iota$ is not a boundary, so C_1 gets a generator $[h_1]$ in degree 2 with $d[h_1] = \mathrm{Sq}^2\iota$. Level 0 gets the element $b_1^0 = (\mathrm{Sq}^2\iota \mid [h_1])$, with leading term $x_{11}\,\iota$. *End of degree 2.* New basis elements: level 0: b_1^0; level 1: b_0^1. New generators: $[h_1] \in C_1$, $[h_0^2] \in C_2$.

Degree $t = 3$

The work of this degree: level 0: pair; level 1: single. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Level 1, single $x_{01} \cdot b_0^1$**.** Since x_{01} divides $\mathrm{LT}(b_0^1) = x_{01}\,[h_0]$ and $x_{in}^2 = 0$ in $E^0\mathcal{A}$, the leading term of $\mathrm{Sq}^1 \cdot b_0^1$ is not the product of the leading terms, and the product has to be reduced:

$$(0 \mid \mathrm{Sq}^1[h_0^2]).$$

The first component is zero, so the second component $z = \mathrm{Sq}^1[h_0^2]$ is a cycle in C_2. No term of it is divisible by a leading term of level 2 from earlier degrees. The cycle $\mathrm{Sq}^1[h_0^2]$ is set aside: cycles are tested once all results of level 1 in this degree are sorted.

The cycles of level 1. There is one. **Outcome: new generator.** The cycle $\mathrm{Sq}^1[h_0^2]$ is not a boundary, so C_3 gets a generator $[h_0^3]$ in degree 3 with $d[h_0^3] = \mathrm{Sq}^1[h_0^2]$. Level 2 gets the element $b_0^2 = (\mathrm{Sq}^1[h_0^2] \mid [h_0^3])$, with leading term $x_{01}\,[h_0^2]$.

Level 0, pair (b_0^0, b_1^0)**.** The leading terms $\mathrm{LT}(b_0^0) = x_{01}\,\iota$ and $\mathrm{LT}(b_1^0) = x_{11}\,\iota$ have least common multiple $x_{01}x_{11}\,\iota$, with cofactors x_{11} and x_{01}. The S-element $\mathrm{Sq}^2 \cdot b_0^0 + \mathrm{Sq}^1 \cdot b_1^0$, in which the leading terms cancel, is

$$(\mathrm{Sq}(0,1)\iota \mid \mathrm{Sq}^1[h_1] + \mathrm{Sq}^2[h_0]).$$

No term of the first component is divisible by a leading term of level 0. **Outcome: new basis element** $b_2^0 = (\mathrm{Sq}(0,1)\iota \mid \mathrm{Sq}^1[h_1] + \mathrm{Sq}^2[h_0])$ of level 0, with leading term $x_{02}\,\iota$. *End of degree 3.* New basis elements: level 0: b_2^0; level 2: b_0^2. New generators: $[h_0^3] \in C_3$.

Degree $t = 4$

The work of this degree: the relation $\mathrm{Sq}^4\iota$; level 0: single, pair; level 2: single. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Level 2, single $x_{01} \cdot b_0^2$**.** Since x_{01} divides $\mathrm{LT}(b_0^2) = x_{01}\,[h_0^2]$ and $x_{in}^2 = 0$ in $E^0\mathcal{A}$, the leading term of $\mathrm{Sq}^1 \cdot b_0^2$ is not the product of the leading terms, and the product has to be reduced:

$$(0 \mid \mathrm{Sq}^1[h_0^3]).$$

The first component is zero, so the second component $z = \mathrm{Sq}^1[h_0^3]$ is a cycle in C_3. There is no level 3 (we stop at $s = 3$). The cycle $\mathrm{Sq}^1[h_0^3]$ is set aside: cycles are tested once all results of level 2 in this degree are sorted.

The cycles of level 2. There is one. **Outcome: beyond.** The cycle $\mathrm{Sq}^1[h_0^3]$ would give a generator of C_4, beyond $s = 3$, so it is dropped.

Level 0, single $x_{11} \cdot b_1^0$. Since x_{11} divides $\mathrm{LT}(b_1^0) = x_{11}\,\iota$ and $x_{in}^2 = 0$ in $E^0\mathcal{A}$, the leading term of $\mathrm{Sq}^2 \cdot b_1^0$ is not the product of the leading terms, and the product has to be reduced:

$$(\mathrm{Sq}(1,1)\iota \mid \mathrm{Sq}^2[h_1]).$$

Reduce the first component modulo level 0:

- The term $\mathrm{Sq}(1,1)\iota$ has leading monomial $x_{01}x_{02}\,\iota$, divisible by $\mathrm{LT}(b_0^0)$ with quotient x_{02}. Adding $\mathrm{Sq}(0,1) \cdot b_0^0 = (\mathrm{Sq}(1,1)\iota \mid \mathrm{Sq}(0,1)[h_0])$ leaves $(0 \mid \mathrm{Sq}^2[h_1] + \mathrm{Sq}(0,1)[h_0])$.

The first component is zero, so the second component $z = \mathrm{Sq}^2[h_1] + \mathrm{Sq}(0,1)[h_0]$ is a cycle in C_1. No term of it is divisible by a leading term of level 1 from earlier degrees. The cycle $\mathrm{Sq}^2[h_1] + \mathrm{Sq}(0,1)[h_0]$ is set aside: cycles are tested once all results of level 0 in this degree are sorted.

Level 0, pair (b_0^0, b_2^0). The leading terms $\mathrm{LT}(b_0^0) = x_{01}\,\iota$ and $\mathrm{LT}(b_2^0) = x_{02}\,\iota$ have least common multiple $x_{01}x_{02}\,\iota$, with cofactors x_{02} and x_{01}. The S-element $\mathrm{Sq}(0,1) \cdot b_0^0 + \mathrm{Sq}^1 \cdot b_2^0$, in which the leading terms cancel, is

$$(0 \mid \mathrm{Sq}^3[h_0] + \mathrm{Sq}(0,1)[h_0]).$$

The first component is zero, so the second component $z = \mathrm{Sq}^3[h_0] + \mathrm{Sq}(0,1)[h_0]$ is a cycle in C_1. Reduce it modulo $\mathrm{im}(d\colon C_2 \to C_1)$, the first components of level 1:

- The term $\mathrm{Sq}^3[h_0]$ has leading monomial $x_{01}x_{11}\,[h_0]$, divisible by $\mathrm{LT}(b_0^1)$ with quotient x_{11}. Adding $\mathrm{Sq}^2 \cdot \pi_1(b_0^1) = \mathrm{Sq}^3[h_0] + \mathrm{Sq}(0,1)[h_0]$ leaves 0.

Outcome: useless. Both components vanish, and the reduction contributed nothing.

The cycles of level 0. There is one. **Outcome: new generator.** The cycle $\mathrm{Sq}^2[h_1] + \mathrm{Sq}(0,1)[h_0]$ is not a boundary, so C_2 gets a generator $[h_1^2]$ in degree 4 with $d[h_1^2] = \mathrm{Sq}^2[h_1] + \mathrm{Sq}(0,1)[h_0]$. Level 1 gets the element $b_1^1 = (\mathrm{Sq}^2[h_1] + \mathrm{Sq}(0,1)[h_0] \mid [h_1^2])$, with leading term $x_{11}\,[h_1]$.

Relation $\mathrm{Sq}^4\iota = 0$. The module relation enters as the element $(0 \mid \mathrm{Sq}^4\iota)$ of level -1, so its C_0 component is a candidate for d of a new generator. It is reduced modulo the image $\mathrm{im}(d\colon C_1 \to C_0)$, i.e. the first components of level 0. No term of it is divisible by a leading term of level 0 from earlier degrees. **Outcome: new generator.** The relation $\mathrm{Sq}^4\iota$ is not a boundary, so C_1 gets a generator $[h_2]$ in degree 4 with $d[h_2] = \mathrm{Sq}^4\iota$. Level 0 gets the element $b_3^0 = (\mathrm{Sq}^4\iota \mid [h_2])$, with leading term $x_{21}\,\iota$. *End of degree 4.* New basis elements: level 0: b_3^0; level 1: b_1^1. New generators: $[h_2] \in C_1$, $[h_1^2] \in C_2$.

Degree $t = 5$

The work of this degree: level 0: pair, pair. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Level 0, pair (b_1^0, b_2^0). The leading terms $\mathrm{LT}(b_1^0) = x_{11}\,\iota$ and $\mathrm{LT}(b_2^0) = x_{02}\,\iota$ have least common multiple $x_{11}x_{02}\,\iota$, with cofactors x_{02} and x_{11}. The S-element $\mathrm{Sq}(0,1)\cdot b_1^0 + \mathrm{Sq}^2 \cdot b_2^0$, in which the leading terms cancel, is

$$(0 \mid \mathrm{Sq}^3[h_1] + \mathrm{Sq}(1,1)[h_0]).$$

The first component is zero, so the second component $z = \mathrm{Sq}^3[h_1] + \mathrm{Sq}(1,1)[h_0]$ is a cycle in C_1. Reduce it modulo $\mathrm{im}(d\colon C_2 \to C_1)$, the first components of level 1:

- The term $\mathrm{Sq}^3[h_1]$ has leading monomial $x_{01}x_{11}\,[h_1]$, divisible by $\mathrm{LT}(b_1^1)$ with quotient x_{01}. Adding $\mathrm{Sq}^1 \cdot \pi_1(b_1^1) = \mathrm{Sq}^3[h_1] + \mathrm{Sq}(1,1)[h_0]$ leaves 0.

Outcome: useless. Both components vanish, and the reduction contributed nothing.

Level 0, pair (b_0^0, b_3^0). The leading terms $\mathrm{LT}(b_0^0) = x_{01}\,\iota$ and $\mathrm{LT}(b_3^0) = x_{21}\,\iota$ have least common multiple $x_{01}x_{21}\,\iota$, with cofactors x_{21} and x_{01}. The S-element $\mathrm{Sq}^4 \cdot b_0^0 + \mathrm{Sq}^1 \cdot b_3^0$, in which the leading terms cancel, is

$$(\mathrm{Sq}(2,1)\iota \mid \mathrm{Sq}^1[h_2] + \mathrm{Sq}^4[h_0]).$$

Reduce the first component modulo level 0:

- The term $\mathrm{Sq}(2,1)\iota$ has leading monomial $x_{11}x_{02}\,\iota$, divisible by $\mathrm{LT}(b_1^0)$ with quotient x_{02}. Adding $\mathrm{Sq}(0,1)\cdot b_1^0 = (\mathrm{Sq}(2,1)\iota \mid \mathrm{Sq}(0,1)[h_1])$ leaves $(0 \mid \mathrm{Sq}^1[h_2] + \mathrm{Sq}(0,1)[h_1] + \mathrm{Sq}^4[h_0])$.

The first component is zero, so the second component $z = \mathrm{Sq}^1[h_2] + \mathrm{Sq}(0,1)[h_1] + \mathrm{Sq}^4[h_0]$ is a cycle in C_1. No term of it is divisible by a leading term of level 1 from earlier degrees. The cycle $\mathrm{Sq}^1[h_2] + \mathrm{Sq}(0,1)[h_1] + \mathrm{Sq}^4[h_0]$ is set aside: cycles are tested once all results of level 0 in this degree are sorted.

The cycles of level 0. There is one. **Outcome: new generator.** The cycle $\mathrm{Sq}^1[h_2] + \mathrm{Sq}(0,1)[h_1] + \mathrm{Sq}^4[h_0]$ is not a boundary, so C_2 gets a generator $[h_0h_2]$ in degree 5 with $d[h_0h_2] = \mathrm{Sq}^1[h_2] + \mathrm{Sq}(0,1)[h_1] + \mathrm{Sq}^4[h_0]$. Level 1 gets the element $b_2^1 = (\mathrm{Sq}^1[h_2] + \mathrm{Sq}(0,1)[h_1] + \mathrm{Sq}^4[h_0] \mid [h_0h_2])$, with leading term $x_{01}\,[h_2]$. *End of degree 5.* New basis elements: level 1: b_2^1. New generators: $[h_0h_2] \in C_2$.

Degree $t = 6$

The work of this degree: level 0: single, pair; level 1: single, single. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Level 1, single $x_{11} \cdot b_1^1$. Since x_{11} divides $\mathrm{LT}(b_1^1) = x_{11}\,[h_1]$ and $x_{in}^2 = 0$ in $E^0\mathcal{A}$, the leading term of $\mathrm{Sq}^2 \cdot b_1^1$ is not the product of the leading terms, and the product has to be reduced:

$$(\mathrm{Sq}(1,1)[h_1] + \mathrm{Sq}(2,1)[h_0] \mid \mathrm{Sq}^2[h_1^2]).$$

No term of the first component is divisible by a leading term of level 1. **Outcome: new basis element** $b_3^1 = (\mathrm{Sq}(1,1)[h_1] + \mathrm{Sq}(2,1)[h_0] \mid \mathrm{Sq}^2[h_1^2])$ of level 1, with leading term $x_{01}x_{02}\,[h_1]$.

Level 1, single $x_{01} \cdot b_2^1$. Since x_{01} divides $\mathrm{LT}(b_2^1) = x_{01}\,[h_2]$ and $x_{in}^2 = 0$ in $E^0\mathcal{A}$, the leading term of $\mathrm{Sq}^1 \cdot b_2^1$ is not the product of the leading terms, and the product has to be reduced:

$$(\mathrm{Sq}(1,1)[h_1] + \mathrm{Sq}^5[h_0] \mid \mathrm{Sq}^1[h_0h_2]).$$

Reduce the first component modulo level 1:

- The term $\mathrm{Sq}^5[h_0]$ has leading monomial $x_{01}x_{21}\,[h_0]$, divisible by $\mathrm{LT}(b_0^1)$ with quotient x_{21}. Adding $\mathrm{Sq}^4 \cdot b_0^1 = (\mathrm{Sq}^5[h_0] + \mathrm{Sq}(2,1)[h_0] \mid \mathrm{Sq}^4[h_0^2])$ leaves $(\mathrm{Sq}(1,1)[h_1] + \mathrm{Sq}(2,1)[h_0] \mid \mathrm{Sq}^1[h_0h_2] + \mathrm{Sq}^4[h_0^2])$.

Its first component contains $\mathrm{LT}(b_3^1)$ of the element b_3^1 just found in this degree; adding it gives $(0 \mid \mathrm{Sq}^1[h_0h_2] + \mathrm{Sq}^2[h_1^2] + \mathrm{Sq}^4[h_0^2])$. The cycle $\mathrm{Sq}^1[h_0h_2] + \mathrm{Sq}^2[h_1^2] + \mathrm{Sq}^4[h_0^2]$ is set aside: cycles are tested once all results of level 1 in this degree are sorted.

The cycles of level 1. There is one. **Outcome: new generator.** The cycle $\mathrm{Sq}^1[h_0h_2] + \mathrm{Sq}^2[h_1^2] + \mathrm{Sq}^4[h_0^2]$ is not a boundary, so C_3 gets a generator $[h_0^2h_2]$ in degree 6 with $d[h_0^2h_2] = \mathrm{Sq}^1[h_0h_2] + \mathrm{Sq}^2[h_1^2] + \mathrm{Sq}^4[h_0^2]$. Level 2 gets the element $b_1^2 = (\mathrm{Sq}^1[h_0h_2] + \mathrm{Sq}^2[h_1^2] + \mathrm{Sq}^4[h_0^2] \mid [h_0^2h_2])$, with leading term $x_{01}\,[h_0h_2]$.

Level 0, single $x_{02} \cdot b_2^0$. Since x_{02} divides $\mathrm{LT}(b_2^0) = x_{02}\,\iota$ and $x_{in}^2 = 0$ in $E^0\mathcal{A}$, the leading term of $\mathrm{Sq}(0,1) \cdot b_2^0$ is not the product of the leading terms, and the product has to be reduced:

$$(0 \mid \mathrm{Sq}(1,1)[h_1] + \mathrm{Sq}(2,1)[h_0]).$$

The first component is zero, so the second component $z = \mathrm{Sq}(1,1)[h_1] + \mathrm{Sq}(2,1)[h_0]$ is a cycle in C_1. No term of it is divisible by a leading term of level 1 from earlier degrees. The second component contains the leading term of b_3^1, found in this degree; adding its first component leaves 0. **Outcome: useless.** The cycle is the first component of b_3^1, found one level up earlier in this degree, so it is the boundary $d(\mathrm{Sq}^2[h_1^2])$. Both components now vanish, and the reduction contributed nothing.

Level 0, pair (b_1^0, b_3^0). The leading terms $\mathrm{LT}(b_1^0) = x_{11}\,\iota$ and $\mathrm{LT}(b_3^0) = x_{21}\,\iota$ have least common multiple $x_{11}x_{21}\,\iota$, with cofactors x_{21} and x_{11}. The S-element $\mathrm{Sq}^4 \cdot b_1^0 + \mathrm{Sq}^2 \cdot b_3^0$, in which the leading terms cancel, is

$$(\mathrm{Sq}(0,2)\iota \mid \mathrm{Sq}^2[h_2] + \mathrm{Sq}^4[h_1]).$$

No term of the first component is divisible by a leading term of level 0. **Outcome: new basis element** $b_4^0 = (\mathrm{Sq}(0,2)\iota \mid \mathrm{Sq}^2[h_2] + \mathrm{Sq}^4[h_1])$ of level 0, with leading term $x_{12}\,\iota$. *End of degree 6.* New basis elements: level 0: b_4^0; level 1: b_3^1; level 2: b_1^2. New generators: $[h_0^2h_2] \in C_3$.

Degree $t = 7$

Level 0 is no longer needed: a cycle found at level s in this degree would be a generator of C_{s+2} in stem $7 - s - 2 > 4$, so its pairs are not formed. The work of this degree: level 1: single; level 2: single. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Level 2, single $x_{01} \cdot b_1^2$. Since x_{01} divides $\mathrm{LT}(b_1^2) = x_{01}\,[h_0h_2]$ and $x_{in}^2 = 0$ in $E^0\mathcal{A}$, the leading term of $\mathrm{Sq}^1 \cdot b_1^2$ is not the product of the leading terms, and the product has to be reduced:

$$(\mathrm{Sq}^3[h_1^2] + \mathrm{Sq}^5[h_0^2] \mid \mathrm{Sq}^1[h_0^2h_2]).$$

Reduce the first component modulo level 2:

- The term $\mathrm{Sq}^5[h_0^2]$ has leading monomial $x_{01}x_{21}\,[h_0^2]$, divisible by $\mathrm{LT}(b_0^2)$ with quotient x_{21}. Adding $\mathrm{Sq}^4 \cdot b_0^2 = (\mathrm{Sq}^5[h_0^2] + \mathrm{Sq}(2,1)[h_0^2] \mid \mathrm{Sq}^4[h_0^3])$ leaves $(\mathrm{Sq}^3[h_1^2] + \mathrm{Sq}(2,1)[h_0^2] \mid \mathrm{Sq}^1[h_0^2h_2] + \mathrm{Sq}^4[h_0^3])$.

Outcome: new basis element $b_2^2 = (\mathrm{Sq}^3[h_1^2] + \mathrm{Sq}(2,1)[h_0^2] \mid \mathrm{Sq}^1[h_0^2h_2] + \mathrm{Sq}^4[h_0^3])$ of level 2, with leading term $x_{01}x_{11}\,[h_1^2]$.

Level 1, single $x_{01} \cdot b_3^1$. Since x_{01} divides $\mathrm{LT}(b_3^1) = x_{01}x_{02}\,[h_1]$ and $x_{in}^2 = 0$ in $E^0\mathcal{A}$, the leading term of $\mathrm{Sq}^1 \cdot b_3^1$ is not the product of the leading terms, and the product has to be reduced:

$$(\mathrm{Sq}(3,1)[h_0] \mid \mathrm{Sq}^3[h_1^2]).$$

Reduce the first component modulo level 1:

- The term $\mathrm{Sq}(3,1)[h_0]$ has leading monomial $x_{01}x_{11}x_{02}\,[h_0]$, divisible by $\mathrm{LT}(b_0^1)$ with quotient $x_{11}x_{02}$. Adding $\mathrm{Sq}(2,1) \cdot b_0^1 = (\mathrm{Sq}(3,1)[h_0] \mid \mathrm{Sq}(2,1)[h_0^2])$ leaves $(0 \mid \mathrm{Sq}^3[h_1^2] + \mathrm{Sq}(2,1)[h_0^2])$.

The first component is zero, so the second component $z = \mathrm{Sq}^3[h_1^2] + \mathrm{Sq}(2,1)[h_0^2]$ is a cycle in C_2. No term of it is divisible by a leading term of level 2 from earlier degrees. The second component contains the leading term of b_2^2, found in this degree; adding its first component leaves 0. **Outcome: useless.** The cycle is the first component of b_2^2, found one level up earlier in this degree, so it is the boundary $d(\mathrm{Sq}^1[h_0^2h_2] + \mathrm{Sq}^4[h_0^3])$. Both components now vanish, and the reduction contributed nothing. *End of degree 7.* New basis elements: level 2: b_2^2. No new generators.

4 The result

Generators and differentials.

generator	s	t	$t-s$	differential
ι	0	0	0	—
$[h_0]$	1	1	0	$\mathrm{Sq}^1\iota$
$[h_1]$	1	2	1	$\mathrm{Sq}^2\iota$
$[h_0^2]$	2	2	0	$\mathrm{Sq}^1[h_0]$
$[h_0^3]$	3	3	0	$\mathrm{Sq}^1[h_0^2]$
$[h_2]$	1	4	3	$\mathrm{Sq}^4\iota$
$[h_1^2]$	2	4	2	$\mathrm{Sq}^2[h_1] + \mathrm{Sq}(0,1)[h_0]$
$[h_0h_2]$	2	5	3	$\mathrm{Sq}^1[h_2] + \mathrm{Sq}(0,1)[h_1] + \mathrm{Sq}^4[h_0]$
$[h_0^2h_2]$	3	6	3	$\mathrm{Sq}^1[h_0h_2] + \mathrm{Sq}^2[h_1^2] + \mathrm{Sq}^4[h_0^2]$

Since the resolution is minimal, the generators are a basis of $\mathrm{Ext}_{\mathcal{A}}^{s,t}(\mathbb{F}_2, \mathbb{F}_2)$ in this range:

$s\backslash t-s$	0	1	2	3	4
3	h_0^3			$h_0^2h_2$	
2	h_0^2		h_1^2	h_0h_2	
1	h_0	h_1		h_2	
0	1				

This is the familiar picture: the h_0-tower in stem 0, h_1 and h_1^2 in stems 1 and 2, and h_2, h_0h_2, $h_0^2h_2 = h_1^3$ in stem 3; stem 4 is empty. The cycle beyond $s = 3$ found in degree 4 is the next class h_0^4 of the tower.

The Gröbner bases.

element	t	leading term	first component x_1	second component x_2
b_0^0	1	$x_{01}\,\iota$	$\mathrm{Sq}^1\iota$	$[h_0]$
b_1^0	2	$x_{11}\,\iota$	$\mathrm{Sq}^2\iota$	$[h_1]$
b_2^0	3	$x_{02}\,\iota$	$\mathrm{Sq}(0,1)\iota$	$\mathrm{Sq}^1[h_1] + \mathrm{Sq}^2[h_0]$
b_3^0	4	$x_{21}\,\iota$	$\mathrm{Sq}^4\iota$	$[h_2]$
b_4^0	6	$x_{12}\,\iota$	$\mathrm{Sq}(0,2)\iota$	$\mathrm{Sq}^2[h_2] + \mathrm{Sq}^4[h_1]$
b_0^1	2	$x_{01}\,[h_0]$	$\mathrm{Sq}^1[h_0]$	$[h_0^2]$
b_1^1	4	$x_{11}\,[h_1]$	$\mathrm{Sq}^2[h_1] + \mathrm{Sq}(0,1)[h_0]$	$[h_1^2]$
b_2^1	5	$x_{01}\,[h_2]$	$\mathrm{Sq}^1[h_2] + \mathrm{Sq}(0,1)[h_1] + \mathrm{Sq}^4[h_0]$	$[h_0h_2]$
b_3^1	6	$x_{01}x_{02}\,[h_1]$	$\mathrm{Sq}(1,1)[h_1] + \mathrm{Sq}(2,1)[h_0]$	$\mathrm{Sq}^2[h_1^2]$
b_0^2	3	$x_{01}\,[h_0^2]$	$\mathrm{Sq}^1[h_0^2]$	$[h_0^3]$
b_1^2	6	$x_{01}\,[h_0h_2]$	$\mathrm{Sq}^1[h_0h_2] + \mathrm{Sq}^2[h_1^2] + \mathrm{Sq}^4[h_0^2]$	$[h_0^2h_2]$
b_2^2	7	$x_{01}x_{11}\,[h_1^2]$	$\mathrm{Sq}^3[h_1^2] + \mathrm{Sq}(2,1)[h_0^2]$	$\mathrm{Sq}^1[h_0^2h_2] + \mathrm{Sq}^4[h_0^3]$

Counters. The statistics the program keeps per level ($s = -1$ are the module relations):

s	formed	divisible	chain	singles	reduced	useless	elements	boundary	gens	beyond	steps
-1	0	0	0	0	3	0	0	0	3	0	0
0	5	0	0	3	8	3	2	0	3	0	4
1	0	0	0	4	4	1	1	0	2	0	2
2	0	0	0	2	2	0	1	0	0	1	1
all	5	0	0	9	17	4	4	0	8	1	7

Here "formed" counts critical pairs, "divisible" and "chain" those removed by the Gebauer–Möller criteria (none in this small range), and "steps" the reduction steps. Of the 17 reductions, 4 were useless and one was beyond the range; the other 12 each produced a basis element or a generator.